Hidden Markov Neural Networks

1. What's the core idea?

One of the biggest challenges in AI right now is Continual Learning — how do we build models that can learn new things without completely forgetting what they learned before? This paper tackles that problem in a really interesting way.Instead of treating neural network weights as fixed numbers, the authors treat them as hidden states inside a Factorial Hidden Markov Model (FHMM). That means the weights are no longer static — they evolve over time through a probabilistic transition process.

To make this mathematically manageable, the paper uses Bayes by Backprop, a form of Sequential Variational Inference, along with Variational DropConnect to stabilize the learning process and encourage structured weight evolution.The result is a network that can continuously adapt to changing environments while gradually reducing reliance on outdated information. In a sense, the model learns not only *what* to remember, but also *what* to forget.

2. Why it works well

What stood out to me most is how the model handles uncertainty.

Most traditional deep learning models are extremely overconfident — they can give a completely wrong answer while still being 99% confident in it. This becomes dangerous in real-world systems where the data distribution changes over time.

In the experiments (like rotating MNIST and evolving two-moons datasets), the HMNN behaves differently. When the environment starts shifting or the model encounters patterns it hasn't fully adapted to yet, its Bayesian credible intervals widen. In other words, the model effectively admits: "I'm not fully sure about this yet."

That uncertainty signal is incredibly valuable because it makes the model more trustworthy in non-stationary environments.

The framework also finds a strong balance between stability(retaining useful past knowledge) and plasticity(adapting to new information). Unlike methods such as Elastic Weight Consolidation (EWC), which can become overly restrictive and resist learning new patterns, the HMNN transition dynamics allow for more graceful forgetting. Older information can fade naturally when it stops being useful.

Future Research Directions

1. Hierarchical Multi-Scale Transition Kernels

Current HMNNs apply a uniform transition kernel across all layers. However, research in Hierarchical Hidden Markov Models and temporal feature hierarchies suggests that neural representations evolve at different timescales depending on their level of abstraction.

Direction: Future work could introduce layer-specific transition kernels with different stability-plasticity dynamics across the network hierarchy. Early layers, which capture general and reusable features such as edges, textures, and shapes, could utilize a Slow-Markov kernel with high stability and slow temporal drift. In contrast, deeper task-specific layers could use a Fast-Markov kernel with higher plasticity to rapidly adapt to distribution shifts.

Such a hierarchical transition structure would create a more biologically inspired continual learning system, mimicking the multi-timescale adaptation observed in the human cortex.

2. Reducing the Computational Cost (Amortized Inference)

A major limitation of Bayesian Neural Networks is the computational overhead associated with maintaining parameter distributions (mean and variance) together with the iterative optimization process required for Sequential Variational Inference (SVI).

Direction: Drawing from Amortized Variational Inference (AVI), future research could replace iterative posterior updates with a Hypernetwork-based inference mechanism. Instead of explicitly optimizing the posterior at every time step, a secondary network could learn to directly parameterize or predict the Markovian transition of the primary network's weights in a single forward pass.

3. Continuous-Time Integration with Structured State-Space Models (S4/Mamba)

The current HMNN framework operates in discrete sequential steps, which limits its applicability in environments where observations arrive irregularly over time.

Direction: By integrating HMNN transition dynamics with Neural Ordinary Differential Equations (Neural ODEs) or modern Structured State-Space Models (SSMs) such as S4 and Mamba, the latent weight evolution could be reformulated as a continuous-time stochastic process.

This would allow the framework to:

- process irregularly sampled sensor streams,
- handle mixed-frequency temporal data,
- provide continuous "any-time" predictions,
- and maintain efficient memory scaling independent of sequence length.

4. Markovian Attention for Transformers

Although HMNNs are currently explored primarily in feed-forward settings, transformers remain the dominant architecture in modern AI systems.

Direction: A particularly high-impact research direction would involve extending Markovian weight evolution into transformer attention mechanisms. Instead of treating attention matrices (Key-Query-Value projections) as static after training, the attention dynamics themselves could evolve as hidden states governed by stochastic Markov transitions.

This could allow transformers to dynamically adapt their internal representations during inference, reducing long-context degradation and “context drift” in non-stationary environments. In Large Language Models (LLMs), such a mechanism could enable:

- adaptive long-context reasoning,
- dynamic internal world-model updates,
- continual conversational adaptation,
- and online memory evolution,

without requiring repeated fine-tuning or retraining.